

NOTATION

Some notations used in this book listed
by chapter and section where treated

Chapter I, Section 1

$\in$	is a member of
$\subset$	is a part of
$\cup$	union
$\cap$	intersection
$\emptyset$	the empty set
$-$	difference of sets
$\{x: P(x)\}$	(also $\{P(x)\}$) the set of x such that $P(x)$
$\{a\}$	the set with sole member a
$\{a, b\}$	the set with members a and b
(a, b)	ordered pair
(a, b, c)	ordered triple
(a, b, c, d)	ordered quadruple
$A \times B$	cartesian product
$f: A \rightarrow B$	f maps (mapping) A into B
$f C$	the restriction of f to C
$f(C)$	the image of C with respect to f
$f^{-1}(D)$	the preimage of D with respect to f
$g \circ f$	the composition of g and f
$\mathfrak{P}(A)$	the set of subsets of A
$\bigcup$	union (for a family of sets)
$\bigcap$	intersection (for a family of sets)

Section 2

$a \cdot b$	(also ab) product of a and b
a^{-1}	inverse of a
$a + b$	sum of a and b
0	neutral element or zero of abelian group
$b - a$	b minus a
$-a$	additive inverse of a
b/a	b divided by a
1	identity element of a field

Section 3

$<$	less than
$\leq$	not greater than
$>$	greater than
$\geq$	not less than
$\max X$	the maximum of the set X
$\min X$	the minimum of the set X
$\sup X$	the supremum of the set X
$\inf X$	the infimum of the set X
a^+	the positive part of a
a^-	the negative part of a
$ a $	the absolute value of a
$]a, b[$	the set of x satisfying $a < x < b$
$[a, b]$	the set of x satisfying $a \leq x \leq b$
$]a, b]$	the set of x satisfying $a < x \leq b$
$[a, b[$	the set of x satisfying $a \leq x < b$
A^+	$\{x \in A, x \geq 0\}$, A an ordered field

Section 4

$\mathbf{R}$	the set of real numbers
$\mathbf{P}$	the set of positive real numbers
$\mathbf{N}$	the set of whole (or natural) numbers
$\langle m, n \rangle$	the set of whole numbers x satisfying $m \leq x \leq n$
$(a_n)_{n \in \mathbf{N}}$	the sequence (a_n) with domain $\mathbf{N}$
$\prod_0^n a_k$	the product of the a_k , k ranging from 0 to n
$\sum_0^n a_k$	the sum of the a_k , k ranging from 0 to n
$\mathbf{Z}$	the set of integers
$(a_n)_{n \in \mathbf{Z}}$	the sequence (a_n) with domain $\mathbf{Z}$
$(a_k)_{k \geq n}$	the sequence (a_k) with domain the set of integers not less than n
$(a_k)_{k \leq n}$	the sequence (a_k) with domain the set of integers not exceeding n
$\mathbf{Q}$	the set of rational numbers

$a^{1/n}$ (also $\sqrt[n]{a}$) the n th root of a
 $\sqrt{a}$ the square root of a

Section 5

$\lim_{n \rightarrow \infty} a_n$ (also $\lim a_n$) the limit of (a_n)
 $\uparrow$ never decreasing
 $\downarrow$ never increasing
 $\log x$ the logarithm of x
 $\exp x$ the value of the exponential function at x

Section 6

$\limsup_{n \rightarrow \infty} a_n$ the limit superior of $(a_n)_{n \in \mathbf{N}}$
 $\liminf_{n \rightarrow \infty} a_n$ the limit inferior of $(a_n)_{n \in \mathbf{N}}$

Chapter II, Section 1

$\mathbf{C}$ the set of complex numbers
 i the so designated complex number
 $\operatorname{Re} a$ the real part of a
 $\operatorname{Im} a$ the imaginary part of a
 $\bar{a}$ the conjugate of a
 $|a|$ the absolute value of a ($a \in \mathbf{C}$)

Section 2

σ the map assigning to $(a, b) \in \mathbf{R} \times \mathbf{R}$ the complex number $a + bi$
 $[a, b]$ the set of complex numbers $a(1 - t) + bt$, $0 \leq t \leq 1$
 $C(a; r)$ the set of complex numbers z satisfying $|z - a| = r$
 $\Delta(a; r)$ the set of complex numbers z satisfying $|z - a| < r$

Section 3

$\lim_{n \rightarrow \infty} a_n = \infty$ the limit of the sequence $(a_n)_{n \in \mathbf{N}}$ is ∞
 $f + g$ the sum of the functions f and g
 fg the product of the functions f and g
 αf the product of α and the function f

Section 4

$\sum_q^\infty a_k$ the infinite series “sigma a_k , k ranging from q to ∞ ”
 $|f|$ the absolute value of the function f

Chapter III, Section 1

$\bar{A}$ the closure (or adherence) of A
 $\operatorname{fr} A$ the frontier of A

Section 4

$\gamma_1 \cdot \gamma_2$ the product of the paths γ_1 and γ_2
 $\overrightarrow{ab}$ the path “ a arrow b ”

Section 5

$B(a; r)$	the ball with center a and radius r
ω_f	the modulus of continuity of f

Section 6

$\hat{\mathbb{C}}$	the extended complex plane
∞	the point infinity (of $\hat{\mathbb{C}}$)
$1/f$	the map assigning $1/f(x)$ to x
$\hat{\mathbb{R}}$	the extended real line
$-\infty$	the element $-\infty$ (of $\hat{\mathbb{R}}$)
$+\infty$	the element $+\infty$ (of $\hat{\mathbb{R}}$)
$\sup_A f$	the supremum of the set of values taken by f on A
$\inf_A f$	the infimum of the set of values taken by f on A
$\max_A f$	the maximum of the set of values taken by f on A
$\min_A f$	the minimum of the set of values taken by f on A

Chapter IV, Section 1

$\operatorname{Re} f$	$\operatorname{Re} \circ f$
$\operatorname{Im} f$	$\operatorname{Im} \circ f$
$f'(a)$	the derivative of f at a
$f_+'(a)$	the right-handed derivative of f at a
$f_-'(a)$	the left-handed derivative of f at a
$\operatorname{sg} a$	the signum of a ($\in \mathbb{C}$)

Section 3

$\Delta(a; +\infty)$	the set of complex numbers z satisfying $ z - a < +\infty$, that is, $\mathbb{C}$
$\exp$	exponential function (with domain $\mathbb{C}$ from this point on)
e^z	" e exponent z " or " e to the z "
$S_a(z)$	the sum $\sum_0^\infty \binom{a}{k} z^k$, $ z < 1$
J_0	the Bessel function of order 0

Section 4

$\cos$	cosine
$\sin$	sine
cis	the function defined in this section and so designated
$\operatorname{Log} a$	the principal logarithm of a
$\operatorname{Arg} a$	the principal argument of a

Section 6

X^2	the cartesian product of X with itself
$f_x(a, b)$	the partial derivative of f with respect to x at (a, b)
$f_y(a, b)$	the partial derivative of f with respect to y at (a, b)
$f_{xx}(a, b)$	the second partial derivative " f sub xx at (a, b) ," etc.
$C'(O)$	the class $C'(O)$
$C''(O)$	the class $C''(O)$

Section 7

Arctan a the principal arctangent of a

Section 8

Δu the Laplacian of u

Chapter V, Section 1

$V[f]$ the total variation of f

$L[\gamma]$ the length of γ

$|P|$ the norm of (the partition) P

$\int_a^b f dg$ the Riemann–Stieltjes integral of f with respect to g (over $[a, b]$)

$\int_\gamma f dz$ the integral of f along γ

Section 2

$d(Y)$ diameter of Y

$[a, b; c, d]$ the rectangle a, b, c, d

Section 3

$\text{dist}(A, B)$ the distance between A and B

$\oint_{C(a; r)} f(z) dz$ the integral of f along $C(a; r)$ positively sensed

Section 5

$O(a; \gamma)$ the order of a with respect to γ , *alias* the topological index of γ at a

Section 8

$\text{int } S$ the set of interior points of S

Section 9

$n(a; f)$ the multiplicity of f at a

$v_f(w)$ the valence of f at w

Chapter VI, Section 1

$\sum_{-\infty}^{+\infty} b_k z^k$ the Laurent pair $(\sum_0^\infty b_k z^k, \sum_1^\infty b_{-k} z^{-k})$

$\sum_{-\infty}^{+\infty} a_k$ the sum $\sum_0^\infty a_k + \sum_1^\infty a_{-k}$

$A(a; r_1, r_2)$ $\{r_1 < |z - a| < r_2\}$ where $a \in \mathbb{C}$ and $0 \leq r_1 < r_2 \leq +\infty$

J_k the Bessel function of order k

Section 2

$\Delta(\infty; r)$ the set of z in $\hat{\mathbb{C}}$ satisfying $|z| > r^{-1}$

Section 3

$\text{res}(a; f)$ the residue of f at a

Section 4

$M(O)$ the set of functions meromorphic on O

$\partial_f(a)$ the divisor of f at a

$f + g$	the sum of the meromorphic functions f and g
fg	the product of the meromorphic functions f and g
$\min\{\varphi, \psi\}$	the minimum of the functions φ and ψ
$\chi(a, b)$	the chordal distance between a and b

Chapter VIII, Prefatory Remarks

$\mathbf{A}(A)$	the set of functions analytic on A
Section 1	
E_p	the Weierstrass primary factor
Section 2	
c_A	the constant with domain A taking the value c
Section 3	
$\prod_0^\infty a_n$	the infinite product whose n th factor is a_n
ζ	the Riemann ζ -function
Γ	the gamma function
Section 4	
$S(f)$	the sum of f
g^+	the composition of x^+ , $x \in \hat{\mathbf{R}}$, with g
g^-	the composition of x^- , $x \in \hat{\mathbf{R}}$, with g
$\sum_{x \in B} f(x)$	the sum of $f B$
$\prod_{a \in A} f(a)$	the product of f
$\prod_{a \in A} f_a$	the product of the functions f_a , a taken over A
Section 5	
$v(w; r)$	the valence at w of $f \Delta(0; r)$, f given
$N(0; r)$	denotes $\int_0^r t^{-1} [v(0; t) - \partial_f^+(0)] dt + \partial_f^+(0) \log r$
$N(\infty; r)$	denotes $\int_0^r t^{-1} [v(\infty; t) + \partial_f^-(0)] dt - \partial_f^-(0) \log r$
Section 6	
$M(r)$	the maximum of the set of values taken by $ f $ on $C(0; r)$, f understood
$\alpha(f)$	the order of f
$\kappa(\partial)$	the exponent of convergence of ∂
σ	the Weierstrass σ -function
ζ	the Weierstrass ζ -function (will be distinguished from Riemann ζ -function by explicit reference)
p	the Weierstrass p -function

Chapter XI, Section 1

$f_z(a), f_{\bar{z}}(a)$	the complex differential coefficients of f at a
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Chapter XII, Section 5 $\|a\|$ the norm of a $N(T)$ the norm of the linear transformation T *Chapter XIII, Section 1* $k(\theta, z)$ the Poisson kernel*Section 4* $\max\{f_1, \dots, f_n\}$ the maximum of the functions $f_1, \dots, f_n$ *Chapter XVI, Section 1* λ the modular function defined in this section